

JENKINS TASK-3

Task 1

Create one Jenkins job using the below code and create three stages:

- stage1: Git clone to download the source code.
- stage2: Sonarqube Integration to check the quality of code.
- stage3: Slack Integration to send the alerts to slack.

URL: <https://github.com/betawins/VProfile-1.git>

Stage:1

- create an instance with 20gb and t2 large
- launch instance and allow port number 8080 in SG
- connect with instance and install java and Jenkins

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

.....|

Continue



Jenkins / All / New Item



New Item

Enter an item name

New_job1

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

And then create a job

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/VProfile-1.git

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Save

Apply

Builds

Today

✓ #1 11:28 AM

Jenkins / New_job1

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

New_job1

Permalinks

🔍 ⚙️ 👤

✎ Add description

Builds:

No builds

Build Queue

No builds in the queue.

Build Executor Status

0/2 ▾

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀️	New_job1	1 min 1 sec #1	N/A	1.2 sec

Icons: S M L

Stage:2

```

[ec2-user@ip-172-31-84-206 ~]$ sudo apt update
[sudo: apt: command not found]
[ec2-user@ip-172-31-84-206 ~]$ sudo yum update -y
[ssh: sudo] command not found
[ec2-user@ip-172-31-84-206 ~]$ sudo yum update
[ssh: sudo] command not found
[ec2-user@ip-172-31-84-206 ~]$ sudo yum update -y
search for Red Hat RHEL6 Livepatch repository
dependencies resolved.
235 KB/s | 29 KB 00:00
nothing to do.
Complete!
[ec2-user@ip-172-31-84-206 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
2025-12-26 11:06:38 -- https://pkg.jenkins.io/redhat-stable/jenkins.repo
2025-12-26 11:06:38 -- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Connecting to pkg.jenkins.io [pkg.jenkins.io] : 146.75.34.133, 2024=443:784:645
Connecting to pkg.jenkins.io [pkg.jenkins.io] [146.75.34.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 192 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'
100%[
https://pkg.jenkins.io/redhat-stable/jenkins.repo
2025-12-26 11:06:38 (9.56 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [192/192]
[ec2-user@ip-172-31-84-206 ~]$ sudo yum -i import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
[ec2-user@ip-172-31-84-206 ~]$ sudo yum upgrade
jenkins-stable
jenkins-stable
Importing GPG key 7a2f7975CA:
GPG key:
GPG key fingerprint: 6346 7827 4584 1704 0346 9072 584b 1007 2299 79CA
GPG key url: https://pkg.jenkins.io/redhat-stable/reposd/reposd.xml.key
in state ok [gpg] y
jenkins-stable
jenkins-stable
427 KB/s | 34 KB 00:00

```

```

Installed size: 93 M
Downloading Packages:
jenkins-2.528.3-1.1.noarch.rpm                                40 MB/s |  91 MB | 00:02

Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Running scriptlet: jenkins-2.528.3-1.1.noarch                1/1
  Installing     : jenkins-2.528.3-1.1.noarch                1/1
  Running scriptlet: jenkins-2.528.3-1.1.noarch                1/1
  Verifying      : jenkins-2.528.3-1.1.noarch                1/1

Installed:
jenkins-2.528.3-1.1.noarch

Complete!
ec2-user@ip-172-31-84-206 ~$ sudo systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
ec2-user@ip-172-31-84-206 ~$ sudo systemctl start jenkins
ec2-user@ip-172-31-84-206 ~$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
Loaded loaded: /usr/lib/systemd/system/jenkins.service; enabled; preset: disabled
Active: active (running) since Fri 2025-12-26 11:08:53 UTC; 43 s ago
Main PID: 26458 (java)
Tasks: 45 (limit: 8237)
Memory: 596.3M
CPU: 24.62ms
Group: jpsvc-alice/jenkins-service
└─[26455 /usr/lib/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Dec 26 11:08:47 ip-172-31-84-206.ec2.internal jenkins[26458]: [INFO] This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Dec 26 11:08:47 ip-172-31-84-206.ec2.internal jenkins[26458]: [INFO]
Dec 26 11:08:47 ip-172-31-84-206.ec2.internal jenkins[26458]: [INFO]
Dec 26 11:08:47 ip-172-31-84-206.ec2.internal jenkins[26458]: [INFO]
Dec 26 11:08:47 ip-172-31-84-206.ec2.internal jenkins[26458]: [INFO]
Dec 26 11:08:47 ip-172-31-84-206.ec2.internal jenkins[26458]: [INFO]
Dec 26 11:08:53 ip-172-31-84-206.ec2.internal jenkins[26458]: 2025-12-26 11:08:53.227400000 [ip=39] INFO Jenkins.InitSelectorRunner$InitSelected: Completed initialization
Dec 26 11:08:53 ip-172-31-84-206.ec2.internal jenkins[26458]: 2025-12-26 11:08:53.251400000 [ip=30] INFO hudson.lifecycle.Lifecycle$onReady: Jenkins is fully up and running
Dec 26 11:08:53 ip-172-31-84-206.ec2.internal jenkins[26458]: 2025-12-26 11:08:53.329400000 [ip=54] INFO hudson.DownloadService$DownloadBehavior: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
Dec 26 11:08:53 ip-172-31-84-206.ec2.internal jenkins[26458]: 2025-12-26 11:08:53.330400000 [ip=56] INFO hudson.util.Retrier$start: Performed the action check update server successfully at the attempt #1
ec2-user@ip-172-31-84-206 ~$

```

[illegible]

Installed SQL server

```

Unpacking libec 8.0.44
Downloading Packages:
(1/6): libcrypt-compat-4.4.33-7.amzn2023.x86_64.rpm
(2/6): mysql-community-common-5.7.44-1.el7.x86_64.rpm
(3/6): ncurses-compat-libs-6.2-4.20200222.amzn2023.0.6.x86_64.rpm
(4/6): mysql-community-libs-5.7.44-1.el7.x86_64.rpm
(5/6): mysql-community-client-5.7.44-1.el7.x86_64.rpm
(6/6): mysql-community-server-5.7.44-1.el7.x86_64.rpm

Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 
  Installing     : mysql-community-common-5.7.44-1.el7.x86_64
  Installing     : mysql-community-libs-5.7.44-1.el7.x86_64
  Running scriptlet: mysql-community-libs-5.7.44-1.el7.x86_64
  Installing     : ncurses-compat-libs-6.2-4.20200222.amzn2023.0.6.x86_64
  Installing     : mysql-community-client-5.7.44-1.el7.x86_64
  Installing     : libcrypt-compat-4.4.33-7.amzn2023.x86_64
  Running scriptlet: mysql-community-server-5.7.44-1.el7.x86_64
  Installing     : mysql-community-server-5.7.44-1.el7.x86_64
  Running scriptlet: mysql-community-server-5.7.44-1.el7.x86_64
/usr/lib/tmpfiles.d/mysql.conf:23: Line references path below legacy directory /var/run/, updating /var/run/mysqlid → /run/mysqlid; please update the tmpfiles.d/ drop-in file accordingly.

  Verifying     : libcrypt-compat-4.4.33-7.amzn2023.x86_64
  Verifying     : ncurses-compat-libs-6.2-4.20200222.amzn2023.0.6.x86_64
  Verifying     : mysql-community-client-5.7.44-1.el7.x86_64
  Verifying     : mysql-community-common-5.7.44-1.el7.x86_64
  Verifying     : mysql-community-libs-5.7.44-1.el7.x86_64
  Verifying     : mysql-community-server-5.7.44-1.el7.x86_64

Installed:
  libcrypt-compat-4.4.33-7.amzn2023.x86_64      mysql-community-client-5.7.44-1.el7.x86_64      mysql-community-common-5.7.44-1.el7.x86_64      mysql-community-libs-5.7.44-1.el7.x86_64      mysql-communi

Complete!
[root@ip-172-31-84-206 ~]# sudo systemctl start mysqld.service
[root@ip-172-31-84-206 ~]# systemctl start mysqld
systemctl enable mysqld
[root@ip-172-31-84-206 ~]# netstat -na | grep 3306
tcp6      0      0 :::3306          :::*              LISTEN
[root@ip-172-31-84-206 ~]# ||

```

```

mysql-community-server-8.0.44-1.el7.x86_64
Complete!
[root@ip-172-31-84-206 ~]# sudo systemctl start mysqld.service
[root@ip-172-31-84-206 ~]# systemctl start mysqld
[root@ip-172-31-84-206 ~]# systemctl enable mysqld
[root@ip-172-31-84-206 ~]# netstat -na | grep 3306
[root@ip-172-31-84-206 ~]# systemctl status mysqld
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
   Drop-In: /etc/systemd/system/mysqld.service.d
            └─override.conf
   Active: active (running) since Fri 2025-12-26 13:28:45 UTC; 1min 29s ago
     Docs: man:mysqld(8)
            http://dev.mysql.com/doc/refman/en/using-systemd.html
   Main PID: 40414 (mysqld)
    Status: "Server is operational"
     Tasks: 36 (limit: 9297)
   Memory: 558.9M
      CPU: 7.905s
   CGroup: /system.slice/mysqld.service
            └─40414 /usr/sbin/mysqld --skip-grant-tables --skip-networking

Dec 26 13:28:29 ip-172-31-84-206.ec2.internal systemd[1]: Starting mysqld.service - MySQL Server...
Dec 26 13:28:45 ip-172-31-84-206.ec2.internal systemd[1]: Started mysqld.service - MySQL Server.
[root@ip-172-31-84-206 ~]# grep 'temporary' /var/log/mysqld.log
[root@ip-172-31-84-206 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.44 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> ||

```

```

2 rows in set (0.00 sec)

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.00 sec)

mysql> CREATE DATABASE sonar CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> GRANT ALL PRIVILEGES ON sonar.* TO 'sonar'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL PRIVILEGES ON sonar.* TO 'sonar'@'%';
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sonar |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql>

```

```

| information_schema |
| mysql |
| performance_schema |
| sonar |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> CREATE USER sonar@localhost IDENTIFIED BY 'Sonar@123';
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE USER sonar@'%' IDENTIFIED BY 'Sonar@123';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL ON sonar.* TO sonar@localhost;
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL ON sonar.* TO sonar@'%';
Query OK, 0 rows affected (0.00 sec)

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sonar |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT User FROM mysql.user;
+-----+
| User |
+-----+
| sonar |
| mysql.session |
| mysql.sys |
| root |
| sonar |
+-----+
5 rows in set (0.00 sec)

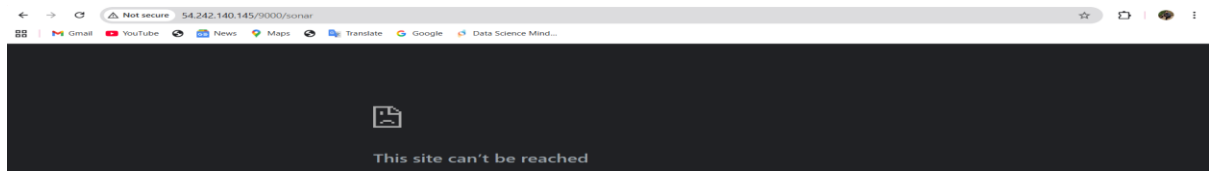
```

```
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> exit
bye
root@ip-172-31-25-121 opt]# vi /opt/sonar/conf/sonar.properties
root@ip-172-31-25-121 opt]# useradd arjumann
root@ip-172-31-25-121 opt]# passwd arjumann
Changing password for user arjumann.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
root@ip-172-31-25-121 opt]# vi /etc/sudoers
root@ip-172-31-25-121 opt]# systemctl restart sshd
root@ip-172-31-25-121 opt]# sudo su arjumann
(arjumann@ip-172-31-25-121 opt)$ cd /opt/sonar/bin/linux-x86-64/
(arjumann@ip-172-31-25-121 linux-x86-64)$ ./sonar.sh start
Starting SonarQube...
failed to start SonarQube.
(arjumann@ip-172-31-25-121 linux-x86-64)$ ./sonar.sh restart
Stopping SonarQube...
SonarQube was not running.
Starting SonarQube...
failed to start SonarQube.
(arjumann@ip-172-31-25-121 linux-x86-64)$ cd /opt/
(arjumann@ip-172-31-25-121 opt)$ ll
total 160500
lrwxr-xr-x. 4 root root          33 Dec  3 16:42 aws
lrwxr-xr-x. 11 root root        141 Oct 20 2017 sonar
-rw-r--r--. 1 root root 164350442 Feb 16 2022 sonarqube-6.6.zip
(arjumann@ip-172-31-25-121 opt)$ sudo chown -R arjumann:arjumann sonar
(arjumann@ip-172-31-25-121 opt]$ cd /sonar/bin/
bash: cd: /sonar/bin/: No such file or directory
(arjumann@ip-172-31-25-121 opt]$ cd cd /opt/sonar/bin/linux-x86-64/
bash: cd: too many arguments
(arjumann@ip-172-31-25-121 opt]$ cd /opt/sonar/bin/linux-x86-64/
(arjumann@ip-172-31-25-121 linux-x86-64)$ ls
lib_sonar.sh  sonarqube
(arjumann@ip-172-31-25-121 linux-x86-64)$ ./sonar.sh start
Starting SonarQube...
Started SonarQube.
(arjumann@ip-172-31-25-121 linux-x86-64)$ ./sonar.sh status
SonarQube is running (29576).
(arjumann@ip-172-31-25-121 linux-x86-64)$ |
```

i-08a21271bc5db85ae (Sonarqube_mysql)
PublicIPs: 54.242.140.145 PrivateIPs: 172.31.25.121

MySQL is not supporting sonarqube it is officially ended



Docker installation

```
(root@ip-172-31-25-121 ~) # yum -y install docker
Last metadata expiration check: 54:52:11 ago on Sun Dec 28 07:12:45 2025.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.13-1.elmn2023.0.2	amazonlinux	46 M
Installing dependencies:				
container-selinux	noarch	4:2.242.0-1.elmn2023	amazonlinux	58 K
iptables-libe	x86_64	2.1.5-1.elmn2023.0.1	amazonlinux	23 M
iptables-nft	x86_64	1.8.8-3.elmn2023.0.2	amazonlinux	401 K
iptables-rtf	x86_64	1.8.8-3.elmn2023.0.2	amazonlinux	183 K
libgroup	x86_64	3.0-1.elmn2023.0.1	amazonlinux	75 K
libnetfilter_conntrack	x86_64	1.0.8-2.elmn2023.0.2	amazonlinux	58 K
libnetfilter_log	x86_64	1.0.1-19.elmn2023.0.2	amazonlinux	50 K
libnftnl	x86_64	1.2.2-2.elmn2023.0.2	amazonlinux	94 K
pkgm	x86_64	2.5-1.elmn2023.0.3	amazonlinux	83 K
runc	x86_64	1.9.3-2.elmn2023.0.1	amazonlinux	3.3 M

```
Transaction Summary
Install 11 Packages

Total download size: 74 M
Installed size: 280 M
Downloading Packages:
(0/11): container-selinux-2.242.0-1.elmn2023.noarch.rpm                                1.4 MB/s | 58 KB  00:00
(2/11): iptables-libe-1.8.8-3.elmn2023.0.2.x86_64.rpm                                9.3 MB/s | 401 KB 00:00
(3/11): iptables-rtf-1.8.8-3.elmn2023.0.2.x86_64.rpm                                4.9 MB/s | 183 KB 00:00
(4/11): libgroup-3.0-1.elmn2023.0.1.x86_64.rpm                                     3.0 MB/s | 75 KB  00:00
(5/11): libnetfilter_conntrack-1.0.8-2.elmn2023.0.2.x86_64.rpm                       2.1 MB/s | 58 KB  00:00
(6/11): libnetfilter_log-1.0.1-19.elmn2023.0.2.x86_64.rpm                           1.0 MB/s | 50 KB  00:00
(7/11): libnftnl-1.2.2-2.elmn2023.0.2.x86_64.rpm                                   1.0 MB/s | 94 KB  00:00
(8/11): pkgm-2.5-1.elmn2023.0.3.x86_64.rpm                                         1.0 MB/s | 83 KB  00:00
(9/11): runc-1.9.3-2.elmn2023.0.1.x86_64.rpm                                       1.0 MB/s | 3.3 MB 00:00
```

Docker pull sonarqube

```
container-selinux-4:2.242.0-1.elmn2023.noarch containerd-2.1.5-1.elmn2023.0.1.x86_64 docker-25.0.13-1.elmn2023.0.2.x86_64 iptables-libe-1.8.8-3.elmn2023.0.2.x86_64 iptables-rtf-1.8.8-3.elmn2023.0.2.x86_64 libgroup-3.0-1.elmn2023.0.1.x86_64 libnetfilter_conntrack-1.0.8-2.elmn2023.0.2.x86_64 libnetfilter_log-1.0.1-19.elmn2023.0.2.x86_64 libnftnl-1.2.2-2.elmn2023.0.2.x86_64 pkgm-2.5-1.elmn2023.0.3.x86_64 runc-1.9.3-2.elmn2023.0.1.x86_64
Complete!
[root@ip-172-31-25-121 ~]# systemctl start docker
Unknown command verb start.
[root@ip-172-31-25-121 ~]# systemctl start docker
[root@ip-172-31-25-121 ~]# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
20043066d3d5: Pull complete
678b78ff6c0: Pull complete
0b82fa77023f: Pull complete
1330e0b0b3e: Pull complete
40e44432c46: Pull complete
f60cf9719a3e: Pull complete
1a022556b0ca: Pull complete
4f4b7b0e754: Pull complete
Digest: sha256:4605079a0c8c777916f781c0f562a0c0b7fe231da95979161a76306043a
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
[root@ip-172-31-25-121 ~]# |
```

i-08a21271bc5db85ae (Sonarqube_mysql)
PublicIPs: 54.242.140.145 PrivateIPs: 172.31.25.121

Docker images

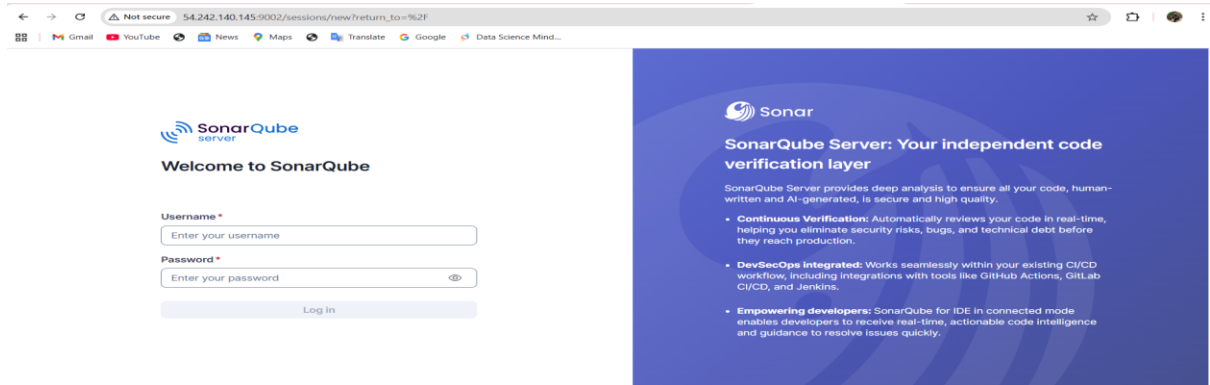
```
sha256:4e55d47ba60c9d70793fa781c0f562a0c39a7fe823da09575141a7636043a
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
[root@ip-172-31-25-121 ~]# docker images
REPOSITORY    ID          IMAGE ID      CREATED      SIZE
sonarqube     latest     2470a055a80b  2 weeks ago  1.27GB
[root@ip-172-31-25-121 ~]#
```

i-08a21271bc5db85ae (Sonarqube_mysql)
PublicPc: 54.242.140.145 PrivatePc: 172.31.25.121

allowed 9002 port number in security group

launch-wizard-9	sg-0ab1db9ac0b46abf7	launch-wizard created 2025-12-28T07:17:01.003Z	vpc-0a2d51a6-4ac8f1fee4
Owner	Inbound rules count	Outbound rules count	
679625722057	3 Permission entries	1 Permission entry	
Inbound rules			
Outbound rules			
Sharing			
VPC associations			
Tags			
Inbound rules (3)			
Search			
☐	Name	Security group rule ID	IP version
☐	-	sg-01ba96278e5f6b17e	IPv4
☐	-	sg-0a5d8955931a2	IPv4
☐	-	sg-0b1ffcb64991f51	IPv4
	Type	Protocol	Port range
	Custom TCP	TCP	9000
	Custom TCP	TCP	9002
	SSH	TCP	22
	Source	Description	
	0.0.0.0/0	-	
	0.0.0.0/0	-	
	0.0.0.0/0	-	

Sonarqube is in deployed successfully



```
Configure here general information about the environment, such as SonarQube server connection details for example
Info: Information about specific project should appear here
#----- Default SonarQube server
#sonar.host.url=http://54.242.140.145:9002/

#----- Default source code encoding
#sonar.sourceEncoding=UTF-8

Last login: Sun Dec 28 08:59:25 2025 from 18.206.107.27
[ec2-user@ip-172-31-21-117 ~]$ sudo su -
Last login: Sun Dec 28 08:39:47 UTC 2025 on pts/1
[root@ip-172-31-21-117 ~]# cd /opt/sonar_scanner
[root@ip-172-31-21-117 sonar_scanner]# cd conf/
[root@ip-172-31-21-117 conf]# ls
sonar-scanner.properties
[root@ip-172-31-21-117 conf]# vi sonar-scanner.properties
[root@ip-172-31-21-117 conf]#
```

▲ Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarQube community Projects Issues Rules Quality Profiles Quality Gates Administration More ▾

A Administrator

Profile Security Notifications Projects

If you want to enforce security by not providing credentials of a real SonarQube user to run your code scan or to invoke web services, you can provide a User Token as a replacement of the user login. This will increase the security of your installation by not letting your analysis user's password going through your network.

Generate Tokens

Name Type Expires in

sonar/-container Global Analysis Token 30 days **Generate**

Name	Type	Project	Last use	Created	Expiration
------	------	---------	----------	---------	------------

Generated key

✓ New token "sonar/-container" has been created. Make sure you copy it now, you won't be able to see it again!

sqa_ce7db2795d5e995989d8df090f1e71721c7e0ada

Security

- Security**
Secure Jenkins; define who is allowed to access/use the system.
- Credentials**
Configure credentials
- Credential Providers**
Configure the credential providers and types
- Users**
Create/delete/modify users that can log in to this Jenkins.

Secret text ▾

Scope ?
Global (Jenkins, nodes, items, all child items, etc) ▾

Secret
.....

ID ?
sonarqube-container

Description ?
sonarqube-container

Create

Install sonaqlube package in plugin

Plugins

- Updates 2
- Available plugins
- Installed plugins
- Advanced settings
- Download progress**

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

SonarQube Scanner

Loading plugin extensions

→ [Go back to the top page](#)
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

To connect to sonarqube I have connected with system configuration added sonarqube credentials

SonarQube installations

List of SonarQube installations

Name

Server URL

Default is <http://localhost:9000>

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

▼ + Add

Advanced ▼

Go to manage Jenkins > tools > add sonar scanner

➤ Add SonarQube Scanner

⚙ SonarQube Scanner

Name

❗ Required

☒ Install automatically ?

⚙ Install from Maven Central

Version

Find the location of the sonar scanner

```
root@ip-172-31-21-117:~#  
Last login: Sun Dec 28 08:59:25 2025 from 18.206.107.27  
(ec2-user@ip-172-31-21-117 ~)$ sudo su -  
Last login: Sun Dec 28 08:59:47 UTC 2025 on pts/1  
[root@ip-172-31-21-117 ~]# cd /opt/sonar_scanner  
[root@ip-172-31-21-117 sonar_scanner]# cd conf/  
[root@ip-172-31-21-117 conf]# ls  
sonar-scanner.properties  
[root@ip-172-31-21-117 conf]# vi sonar-scanner.properties  
[root@ip-172-31-21-117 conf]# cd /opt/sonar_scanner/  
[root@ip-172-31-21-117 sonar_scanner]# ls  
bin conf lib  
[root@ip-172-31-21-117 sonar_scanner]# cd bin/  
[root@ip-172-31-21-117 bin]# ls  
sonar-scanner sonar-scanner-dbg  
[root@ip-172-31-21-117 bin]# pwd  
/opt/sonar_scanner/bin  
[root@ip-172-31-21-117 bin]# ||
```

⚙ SonarQube Scanner

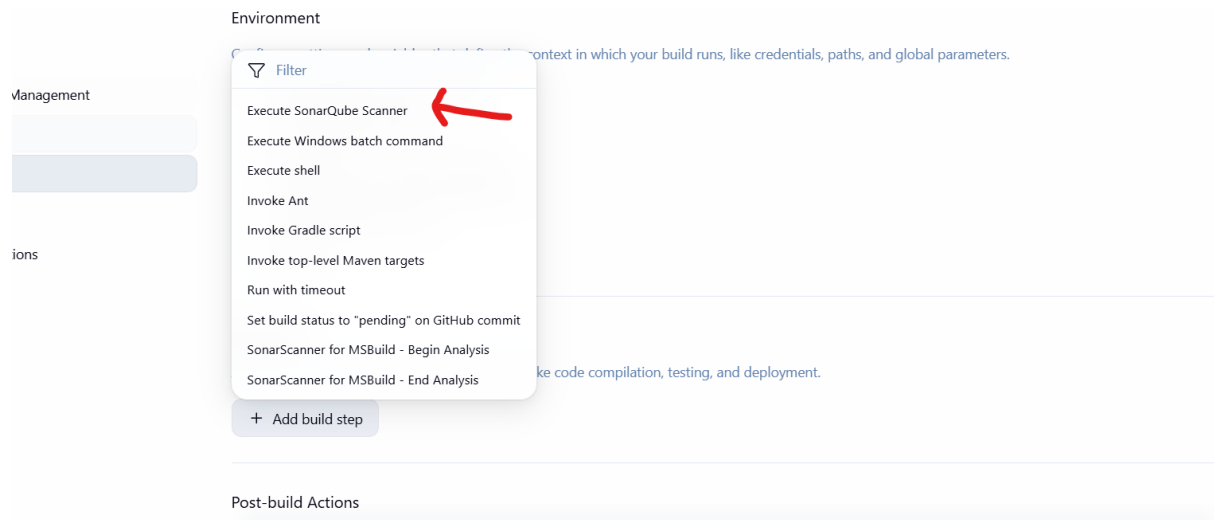
Name

SONAR_RUNNER_HOME

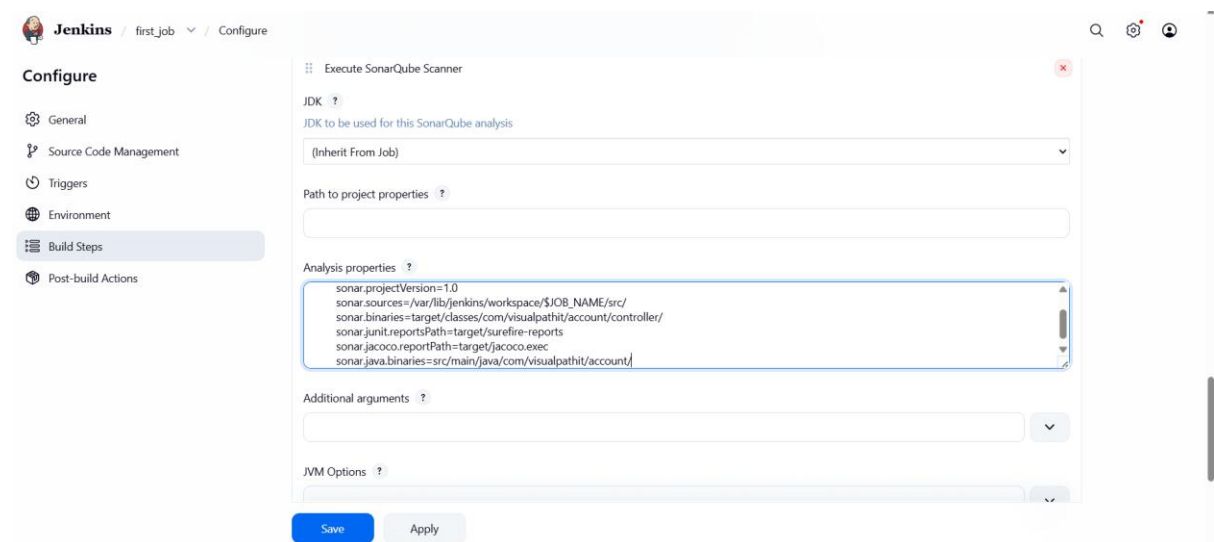
☐ Install automatically ?

+ Add SonarQube Scanner

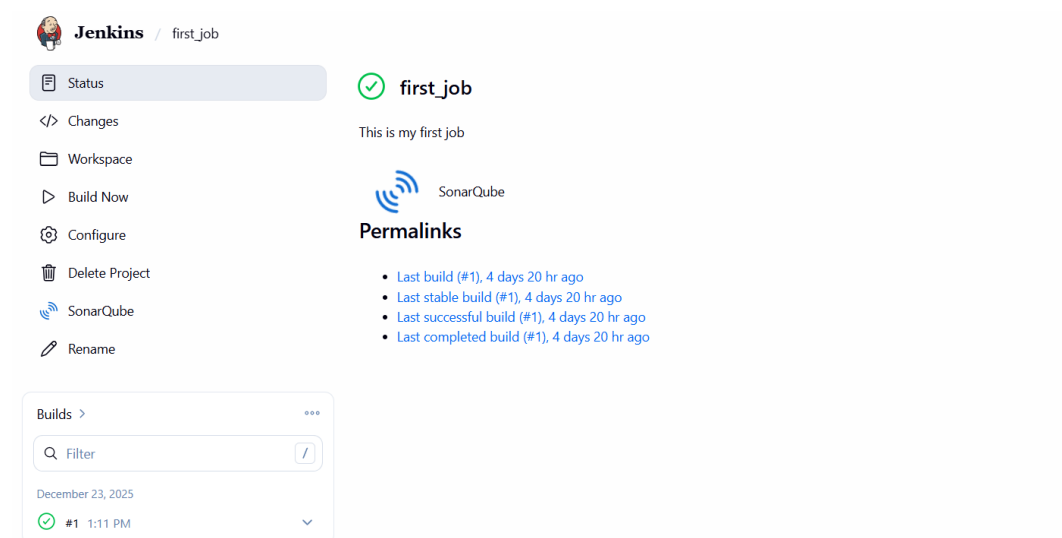
Ant installations



Make changes in the job and add properties in SonarQube section



Logo was appeared as sonar



Job was executed and successfully it is updated in sonarqube server

Job is executed successfully

The screenshot shows the Jenkins job execution details for job #5. The job is in a 'Completed' state, indicated by a green checkmark. The build was started by user 'Ajumand Mohiuddin' on December 28, 2025, at 10:21:04 AM. The build duration was 12 seconds. The build was successful, with no changes detected. The build was triggered by a GitHub repository: <https://github.com/betawins/VProfile-1.git>. The build was triggered by a GitHub repository: <https://github.com/betawins/VProfile-1.git>. The build was triggered by a GitHub repository: <https://github.com/betawins/VProfile-1.git>.

Code is successfully send to sonarqube and tested the quality of the code

The screenshot shows the SonarQube project status for 'Sabear'. The project is in a 'Passed' state, indicated by a green checkmark. The last analysis was performed 45 seconds ago. The project has 17 bugs, 0 vulnerabilities, 0.0% hotspots reviewed, 8 code smells, and 19.4% duplications. The project has 1.2k lines of code. The project is analyzed by JSP, CSS, and other languages.

Stage 3 Configure with slag

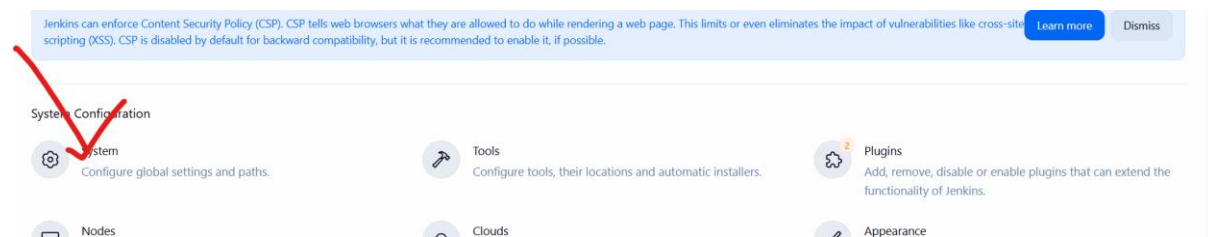
Go to slack click create new and select app select Jenkins ci

The screenshot shows the Slack app selection screen. The 'Jenkins CI' app is highlighted, indicating it is the selected app for integration. The app is described as 'An open source continuous integration server.'.

Go to manage Jenkins select plugin and select slack notification then install it

The screenshot shows the Jenkins plugin management screen. The 'Slack Notification' plugin is selected, and the 'Global Slack Notifier' plugin is also visible. The 'Slack Notification' plugin is described as 'Integrates Jenkins with Slack, allows publishing build statuses, messages and files to Slack channels.'.

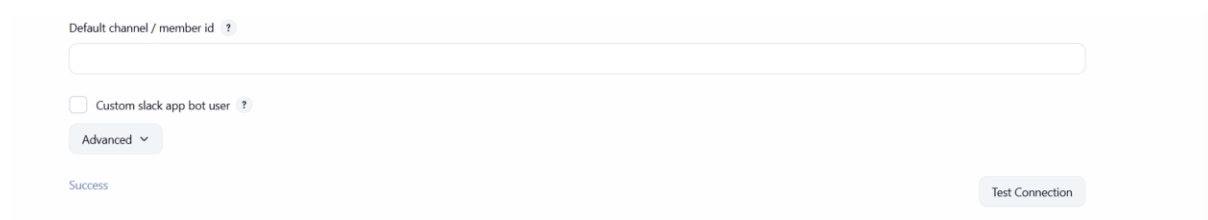
Go to manage Jenkins > system



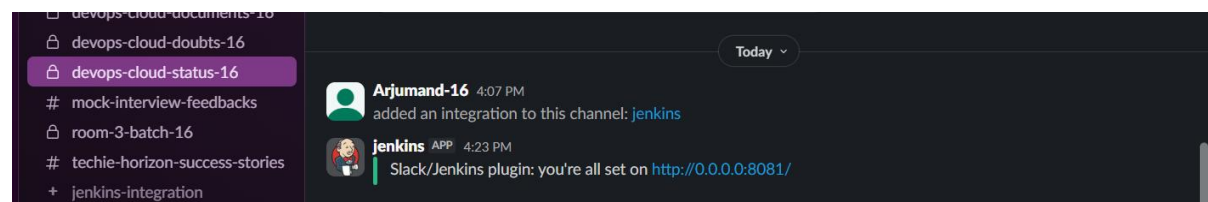
Select slack selection add name of the slack organisation and token number provided by slack which will be added in credentials.



After adding credentials go for test connection



Slack is integrated sucessfully



Task 2

Create one Jenkins job using the below code and create three stages:

- stage1: Git clone to download the source code.
- stage2: Sonarqube Integration to check the quality of code.
- stage3: Slack Integration to send the alerts to slack.

Clone was done using new url- <https://github.com/betawins/hiring-app.git> with the main branch, instead of master.

- stage1: Git clone to download the source code.

The image shows the Jenkins web interface for a build named 'postgresql-job' (build #2). The build status is 'Success' (green checkmark). The build was started by user 'Ajumand Mohiuddin' on Dec 29, 2025, at 9:40:34 AM. The build log shows the following steps: 'Started by user Ajumand Mohiuddin', 'This run spent: 2 ms waiting, 0.1 sec build duration, 0.1 sec total from scheduled to completion.', and 'Revision: 9405b107cfd68e69fe953ea1f9669784f9d6a'. The repository is 'https://github.com/betawins/hiring-app.git'.

Clone is successful as we can see in the terminal as well

```
[ec2-user@ip-172-31-21-117 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-21-117 ~]$ ls
jenkins_backup.sh  test.py
[ec2-user@ip-172-31-21-117 ~]$ cd /var/lib/jenkins/jobs/
[ec2-user@ip-172-31-21-117 jobs]$ ls
Sample_job1  first_job  github-private-repo  nwe  postgresql-job  sample_job3  v-profile-app  webhook_job
[ec2-user@ip-172-31-21-117 jobs]$
```

- Stage2: Sonarqube Integration to check the quality of code.

Installed new server for sonarqube using postgresql using command 'yum install postgresql115-server postgresql115 -y'

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-27-28 ~]$ sudo su -
[root@ip-172-31-27-28 ~]# sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-27-28 ~]# yum install postgresql115-server postgresql115 -y
All metadata expiration checks: 0100125 ago on Mon Dec 29 07:04:18 2025.
dependencies resolved.

Package Architecture Version Repository Size
Installing:
postgresql115-server x86_64 15.15-1.amzn2023.0.1 amazonlinux 6.3 M
installing dependencies:
libicu x86_64 67.1-7.amzn2023.0.4 amazonlinux 9.4 M
postgresql115 x86_64 15.15-1.amzn2023.0.1 amazonlinux 1.7 M
postgresql115-private-liba x86_64 15.15-1.amzn2023.0.1 amazonlinux 143 k

Transaction Summary
Install 4 Packages
Total download size: 18 M
Installed size: 44 M
Downloading Packages:
1/4: postgresql115-private-liba-15.15-1.amzn2023.0.1.x86_64.rpm 2.0 MB/s | 145 kB 00:00
2/4: postgresql115-15.15-1.amzn2023.0.1.x86_64.rpm 16 MB/s | 1.7 MB 00:00
3/4: libicu-67.1-7.amzn2023.0.4.x86_64.rpm 35 MB/s | 9.4 MB 00:00
4/4: postgresql115-server-15.15-1.amzn2023.0.1.x86_64.rpm 25 MB/s | 6.3 MB 00:00
```

sudo /usr/bin/postgresql-setup--initdb--unit postgresql

Now enable and start PostgreSQL:

bash

Copy code

sudo systemctl enable postgresql

sudo systemctl start postgresql

Check status:

bash

Copy code

sudo systemctl status postgresql



Check PostgreSQL Login

Bash

Copy code

sudo-u postgres psql

```
[root@localhost ~]# sudo yum install postgresql
Total download size: 10 M
Installed size: 66 M
Downloading Packages:
(1/4): postgresql15-private-libe-15.15-1.amzn2023.0.1.x86_64.rpm                2.0 MB/s | 14
(2/4): postgresql15-15.15-1.amzn2023.0.1.x86_64.rpm                          16 MB/s | 12
(3/4): libice-1:1.7.amzn2023.0.4.x86_64.rpm                                   35 MB/s | 9.7
(4/4): postgresql15-server-15.15-1.amzn2023.0.1.x86_64.rpm                   25 MB/s | 6.2
Total: 49 MB/s | 10
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 
  Installing     : postgresql15-private-libe-15.15-1.amzn2023.0.1.x86_64
  Installing     : postgresql15-15.15-1.amzn2023.0.1.x86_64
  Installing     : libice-1:1.7.amzn2023.0.4.x86_64
  Running scriptlet: postgresql15-server-15.15-1.amzn2023.0.1.x86_64
  Installing     : postgresql15-server-15.15-1.amzn2023.0.1.x86_64
  Running scriptlet: postgresql15-server-15.15-1.amzn2023.0.1.x86_64
  Verifying      : libice-1:1.7.amzn2023.0.4.x86_64
  Verifying      : postgresql15-15.15-1.amzn2023.0.1.x86_64
  Verifying      : postgresql15-private-libe-15.15-1.amzn2023.0.1.x86_64
  Verifying      : postgresql15-server-15.15-1.amzn2023.0.1.x86_64
Installed:
  libice-1:1.7.amzn2023.0.4.x86_64                postgresql15-15.15-1.amzn2023.0.1.x86_64                postgresql15-private-libe-15.15-1.amzn2023.0.1.x86_64                postgresql15-server-15.15-1.amzn2023.0.1.x86_64
Complete!
[root@ip-172-31-27-28 ~]# /usr/pgsql-15/bin/postgresql-15-setup initdb
bash: /usr/pgsql-15/bin/postgresql-15-setup: No such file or directory
[root@ip-172-31-27-28 ~]# /usr/bin/postgresql-setup --initdb --unit postgresql
* Initializing database in /usr/lib/pgsql/data
* Initializing logs in /usr/lib/pgsql/initdb-postgresql.log
[root@ip-172-31-27-28 ~]# systemctl enable postgresql-15
Failed to enable unit: Unit file postgresql-15.service does not exist.
[root@ip-172-31-27-28 ~]# sudo systemctl enable postgresql
Created symlink /etc/systemd/system/multi-user.target.wants/postgresql.service - /usr/lib/systemd/system/postgresql.service.
[root@ip-172-31-27-28 ~]# sudo systemctl start postgresql
[root@ip-172-31-27-28 ~]# sudo systemctl status postgresql
PostgreSQL Service - PostgreSQL database server
Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: disabled)
Active: active (running) since Mon 2023-12-28 07:07:52 UTC; 10s ago
Process: 27780 ExecStartPre=/usr/libexec/postgresql-check-db-dir postgresql (code=exited, status=0/SUCCESS)
```

Run these commands one-by-one inside the PostgreSQL prompt:

CREATE DATABASE sonarqube;

CREATE USER sonar WITH ENCRYPTED PASSWORD 'StrongPassword123';

GRANT ALL PRIVILEGES ON DATABASE sonarqube TO sonar;

ALTER USER sonar WITH SUPERUSER;

(You can replace password with your own secure one)

Exit PostgreSQL:

\q

```
Dec 29 07:07:57 ip-172-31-27-28.ec2.internal systemd[1]: Starting postgresql.service - PostgreSQL database server...
Dec 29 07:07:57 ip-172-31-27-28.ec2.internal postgres[27780]: 2023-12-29 07:07:57.686 UTC [27780] LOG: redirecting log output to logging collector process
Dec 29 07:07:57 ip-172-31-27-28.ec2.internal postgres[27780]: 2023-12-29 07:07:57.686 UTC [27780] HINT: Future log output will appear in directory "/log".
Dec 29 07:07:57 ip-172-31-27-28.ec2.internal systemd[1]: Started postgresql.service - PostgreSQL database server.
[root@ip-172-31-27-28 ~]# sudo -u postgres psql
psql (15.15)
Type "help" for help.

postgres=# CREATE DATABASE sonarqube;
CREATE DATABASE
postgres=# CREATE USER sonar WITH PASSWORD 'StrongPass123';
CREATE ROLE
postgres=# GRANT ALL PRIVILEGES ON DATABASE sonarqube TO sonar;
GRANT
postgres=# ALTER USER sonar WITH SUPERUSER;
ALTER ROLE
postgres=# \q
[root@ip-172-31-27-28 ~]#
```

Configure PostgreSQL to Allow Local Authentication

Open pg_hba.conf:

sudo nano /var/lib/pgsql/data/pg_hba.conf

Change these lines (at bottom usually):

IPv4 local connections:

```
host      replication    all          ::1/128      ident
# IPv4 local connections:
host      all          all          127.0.0.1/32 md5
:wq
```

```
ident_file = 'ConfigDir/pg_ident.conf' # ident configuration file
#                                     # (change requires restart)

# If external pid file is not explicitly set, no extra PID file is written.
#                                     # write an extra PID file
#external_pid_file = ''                # (change requires restart)

#-----
# CONNECTIONS AND AUTHENTICATION
#-----

# - Connection Settings -

listen_addresses = '*'                # what IP address(es) to listen on;
#                                     # comma-separated list of addresses;
#                                     # defaults to 'localhost'; use '*' for all
#                                     # (change requires restart)
#port = 5432                           # (change requires restart)
max_connections = 100                  # (change requires restart)
#superuser_reserved_connections = 3    # (change requires restart)
-- INSERT --
```

```
[root@ip-172-31-27-28 data]# yum install amazon-linux-extras install java-openjdk17 -y
Last metadata expiration check: 0:29:32 ago on Mon Dec 29 07:04:18 2025.
No match for argument: amazon-linux-extras
No match for argument: install.
No match for argument: java-openjdk17
Error: Unable to find a match: amazon-linux-extras install java-openjdk17
[root@ip-172-31-27-28 data]# sudo dnf install java-17-amazon-corretto -y
Last metadata expiration check: 0:32:20 ago on Mon Dec 29 07:04:18 2025.
Dependencies resolved.
```

Package	Architecture	Version	Repository
---------	--------------	---------	------------

- ```
complete!
[root@ip-172-31-27-28 data]# cd /opt
[root@ip-172-31-27-28 opt]# sudo wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.9.1.69595.zip
--2025-12-29 07:16:27-- https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.9.1.69595.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 99.84.186.21, 99.84.186.106, 99.84.186.35, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com):80. 99.84.186.21:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 293875683 (280K) [binary/octet-stream]
Saving to: 'sonarqube-9.9.1.69595.zip'
```

```
creating: sonarqube-9.9.1.69595/elasticsearch/plugins/
[root@ip-172-31-27-28 opt]# ls
aws_sonarqube-9.9.1.69595 sonarqube-9.9.1.69595.zip
[root@ip-172-31-27-28 opt]# sudo mv sonarqube-9.9.1.69595 sonarqube
[root@ip-172-31-27-28 opt]# ls
aws_sonarqube sonarqube-9.9.1.69595.zip
[root@ip-172-31-27-28 opt]#
```

- `sudo groupadd sonar`
- `sudo useradd -d /opt/sonarqube -g sonar sonar`

➤ `sudo chown -R sonar:sonar /opt/sonarqube`

```
[root@ip-172-31-27-28 opt]# sudo groupadd sonar
sudo useradd -d /opt/sonarqube -g sonar sonar
sudo chown -R sonar:sonar /opt/sonarqube
useradd: warning: the home directory /opt/sonarqube already exists.
useradd: Not copying any file from skel directory into it.
[root@ip-172-31-27-28 opt]# ll
total 286988
drwxr-xr-x. 4 root root 33 Dec 3 16:42 aws
drwxr-xr-x. 11 sonar sonar 172 Apr 27 2023 sonarqube
-rw-r--r--. 1 root root 293875683 Apr 28 2023 sonarqube-9.9.1.69595.zip
[root@ip-172-31-27-28 opt]# ||
```

## Edit config:

`sudo nano /opt/sonarqube/conf/sonar.properties`

Set:

`sonar.jdbc.username=sonar`

`sonar.jdbc.password=StrongPass123`

`sonar.jdbc.url=jdbc:postgresql://localhost:5432/sonarqube`

Enable:

`sonar.web.port=9000`

Save and close: CTRL+O → ENTER → CTRL+X

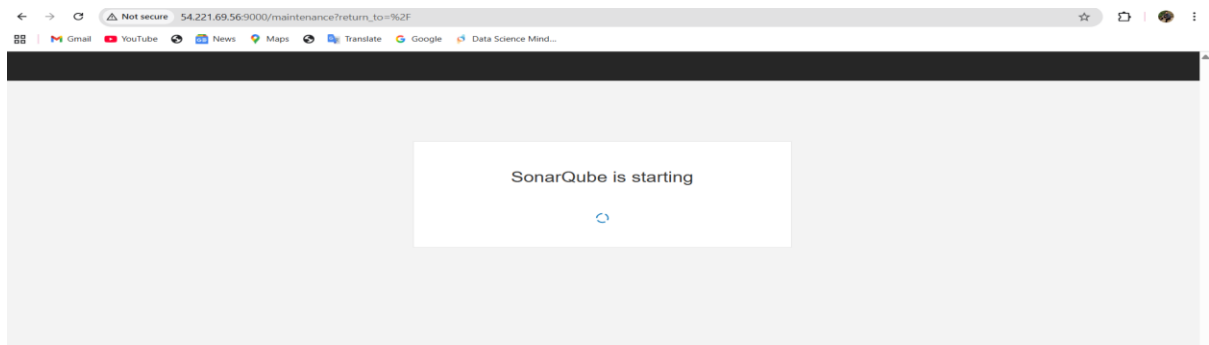
```
User credentials.
Permissions to create tables, indices and triggers must be granted to JDBC user.
The schema must be created first.
sonar.jdbc.username=sonar
sonar.jdbc.password=Strong123
|
#----- Embedded Database (default)
If embedded database server listening port, defaults to 9092
```

```
#----- PostgreSQL 11 or greater
By default the schema named "public" is used. It can be overridden with the parameter "currentSchema".
sonar.jdbc.url=jdbc:postgresql://localhost:5432/sonarqube
```

```
sonar.web.host=0.0.0.0
```

```
Web context. When set, it must start with forward slash (for example /sonarqube).
Changing this value and restarting the server can have unexpected consequences for logged in users,
as they may end up with multiple conflicting browser cookies. If your users experience odd
behaviors after you changed this value and restarted the server, ask them to clear their browser's
cookies and log in again.
The default value is root context (empty value).
#sonar.web.context=
TCP port for incoming HTTP connections. Default value is 9000.
sonar.web.port=9000
```

## Now port is working on 9000 allowed in SG





## **ERROR: I was unable to start sonarqube and deploy on website, I was getting This error after the setup of the steps:**

Since SonarQube is now **running successfully**, here is a clean and professional summary of the **exact troubleshooting steps** I performed.

---

### **SonarQube Deployment Troubleshooting – Summary**

While deploying SonarQube on EC2 with PostgreSQL, SonarQube initially failed to start and Web UI was not accessible. Here are the steps taken to fix the issues:

---

#### **1** Fix Linux Kernel Limits for Elasticsearch

SonarQube requires higher vm.max\_map\_count for Elasticsearch:

```
sudo sysctl-w vm.max_map_count=262144
```

To make it permanent:

```
echo "vm.max_map_count=262144" | sudo tee-a /etc/sysctl.conf
```

```
sudo sysctl-p
```

---

#### **2** Set Correct Ownership and Permissions

Elasticsearch cannot run as root. So we created sonar user and assigned permissions:

```
sudo useradd-d /opt/sonarqube-g sonar sonar
```

```
sudo chown-R sonar:sonar /opt/sonarqube
```

```
sudo chmod-R 775 /opt/sonarqube
```

Then started SonarQube as sonar user:

```
sudo su- sonar
```

```
cd /opt/sonarqube/bin/linux-x86_64/
```

```
./sonar.sh start
```

---

#### **3** Correct PostgreSQL Authentication Errors

SonarQube logs showed:

FATAL: Ident authentication failed for user "sonar"

So I updated PostgreSQL authentication mode from ident to md5:

Edited file:

```
/var/lib/pgsql/15/data/pg_hba.conf
```

Updated rules:

```
host all all 127.0.0.1/32 md5
```

```
host all all ::1/128 md5
```

Restarted PostgreSQL:

```
sudo systemctl restart postgresql
```

---

### Created SonarQube Database and Password

Logged into PostgreSQL:

```
sudo su- postgres
```

```
psql
```

Executed:

```
ALTER USER sonar WITH PASSWORD 'Sonar123';
```

```
CREATE DATABASE sonarqube OWNER sonar;
```

```
\q
```

Tested connection:

```
psql-U sonar-d sonarqube-h 127.0.0.1-W
```

---

### Updated SonarQube Configuration

Edited:

```
/opt/sonarqube/conf/sonar.properties
```

Configured database URL and credentials:

```
sonar.jdbc.username=sonar
```

```
sonar.jdbc.password=Sonar123
```

```
sonar.jdbc.url=jdbc:postgresql://localhost:5432/sonarqube
```

---

## Opened Required Port in AWS

Inbound rules updated for EC2 security group:

| Port | Protocol | Source |
|------|----------|--------|
|------|----------|--------|

|      |     |           |
|------|-----|-----------|
| 9000 | TCP | 0.0.0.0/0 |
|------|-----|-----------|

|    |     |           |
|----|-----|-----------|
| 22 | TCP | 0.0.0.0/0 |
|----|-----|-----------|

---

## Restarted SonarQube & Verified

```
sudo su- sonar
```

```
cd /opt/sonarqube/bin/linux-x86_64/
```

```
./sonar.sh stop
```

```
./sonar.sh start
```

```
./sonar.sh status
```

Checked logs until no errors:

```
tail-f /opt/sonarqube/logs/sonar.log
```

---

## Final Result

SonarQube is successfully deployed and accessible at:

<http://<Public-IP>:9000>

Login:

username: admin

password: admin

---

## Conclusion

By fixing:

✓ Kernel settings

✓ File permissions

✓ PostgreSQL authentication

- ✓ Database configuration
- ✓ Security group access

SonarQube started successfully and UI became accessible.

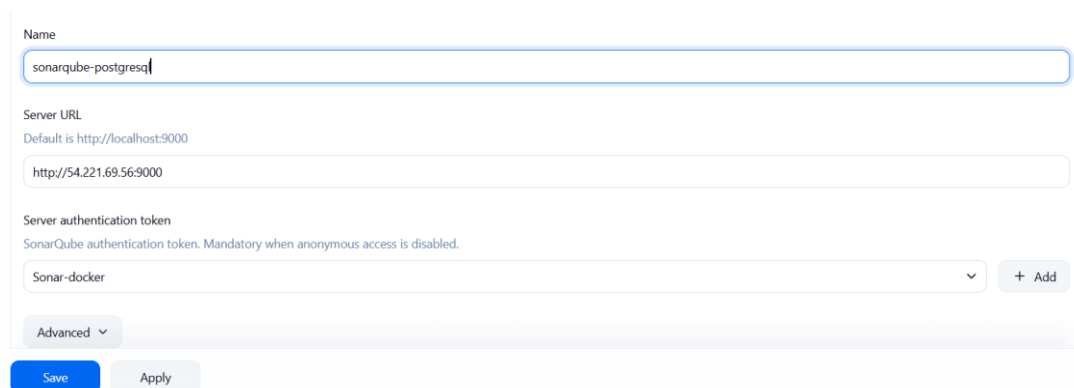
Would you like me to create your Jenkins stages now? 🚀

Updated password:



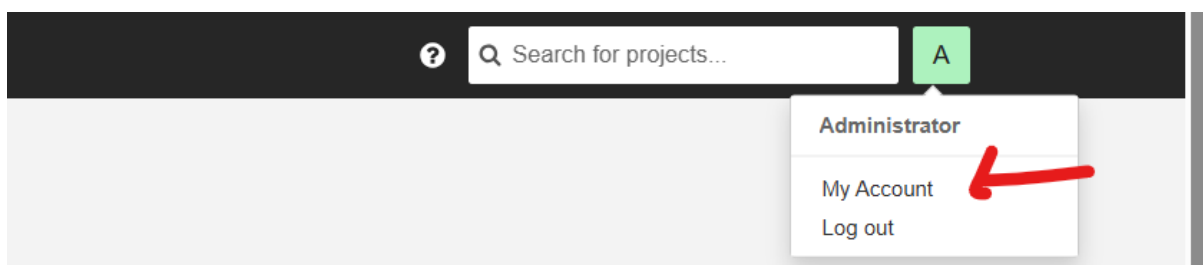
The screenshot shows the 'Update your password' form in SonarQube. The title is 'Update your password'. Below it, a message states: 'This account should not use the default password.' The form asks to 'Enter a new password' and notes that 'All fields marked with \* are required'. There are three input fields: 'Old Password \*', 'New Password \*', and 'Confirm Password \*'. Each field contains a series of dots representing masked text. At the bottom of the form is an 'Update' button.

In Jenkins I have added new url under sonarqube installation



The screenshot shows the Jenkins configuration page for SonarQube. The 'Name' field is filled with 'sonarqube-postgresq'. The 'Server URL' field is filled with 'http://54.221.69.56:9000'. The 'Server authentication token' dropdown is set to 'Sonar-docker'. There are 'Save' and 'Apply' buttons at the bottom.

Go to Soanrqube server and go to logo and choose my account and the go to security tab



Enter the details in security tab and create token which will be added in system configuration in jenkins

A
Administrator

Profile
Security
Notifications
Projects

Tokens

If you want to enforce security by not providing credentials of a real SonarQube user to run your code scan or to invoke web services, you can provide a User Token as a replacement of the user login. This will increase the security of your installation by not letting your analysis user's password going through your network.

Generate Tokens

Name
Type
Expires in

Select Token Type

30 days

Generate

| Name      | Type | Project | Last use | Created | Expiration |
|-----------|------|---------|----------|---------|------------|
| No tokens |      |         |          |         |            |

Add sonarqube token in credentials in secret text option. And provide name and description

Name

sonarqube-postgresql

Server URL

Default is http://localhost:9000

http://54.221.69.56:9000

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

jenkins-sonarqube

+ Add

Advanced

Save
Apply

In Jenkins server add new url of sonarqube url after installing in opt, path will be /opt/sonar\_scannar/conf/sonar-scannar.properties

```
#Configure here general information about the environment, such as SonarQube server connection details for example
#No information about specific project should appear here

#----- Default SonarQube server
sonar.host.url=http://http://54.221.69.56:9000

#----- Default source code encoding
#sonar.sourceEncoding=UTF-8
```

Go to tools in Jenkins and pass the sonar scanner path and name

Jenkins
Manage Jenkins
Tools

SonarScanner for MSBuild

Name

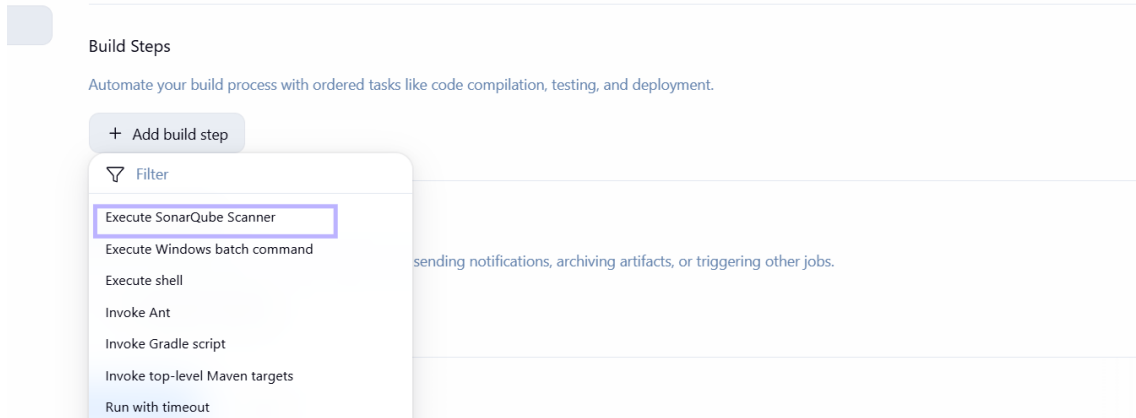
sonar\_scannar

MSBUILD\_SQ\_SCANNER\_HOME

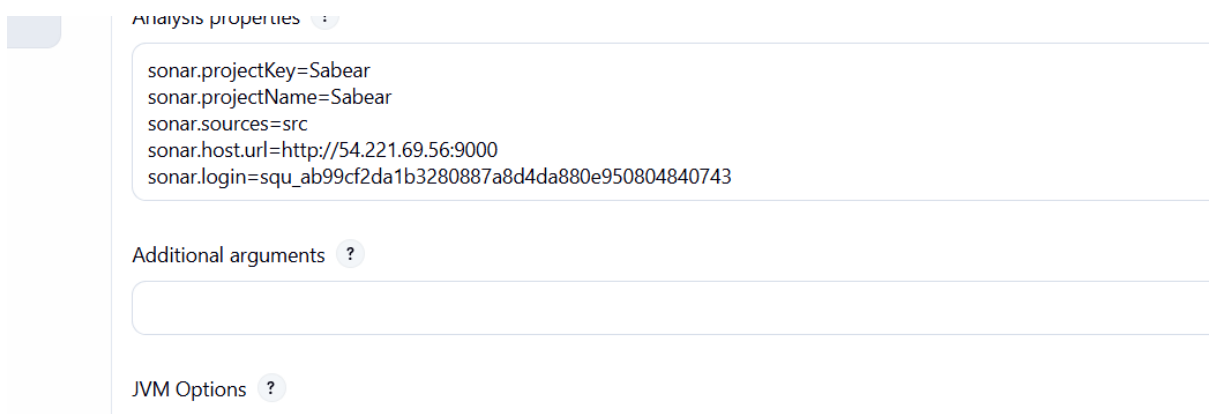
/opt/sonar\_scanner

☐ Install automatically

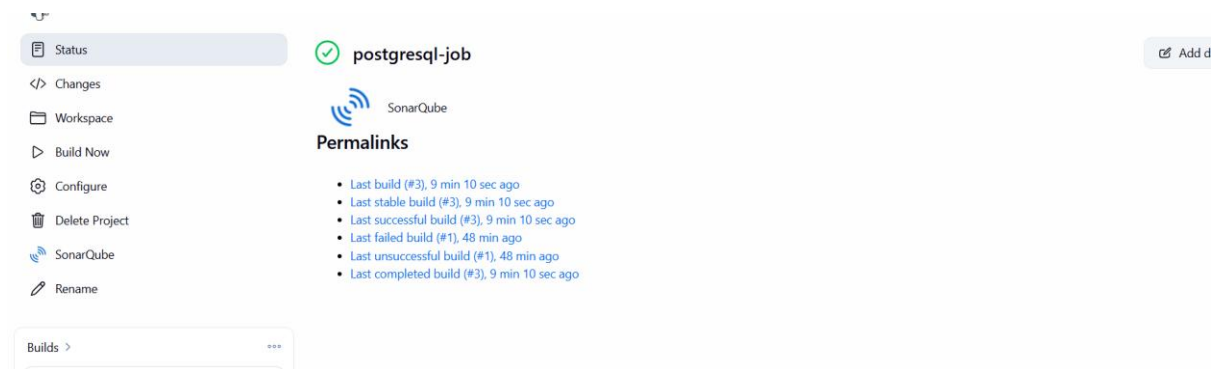
Go to jobs and buit steps add



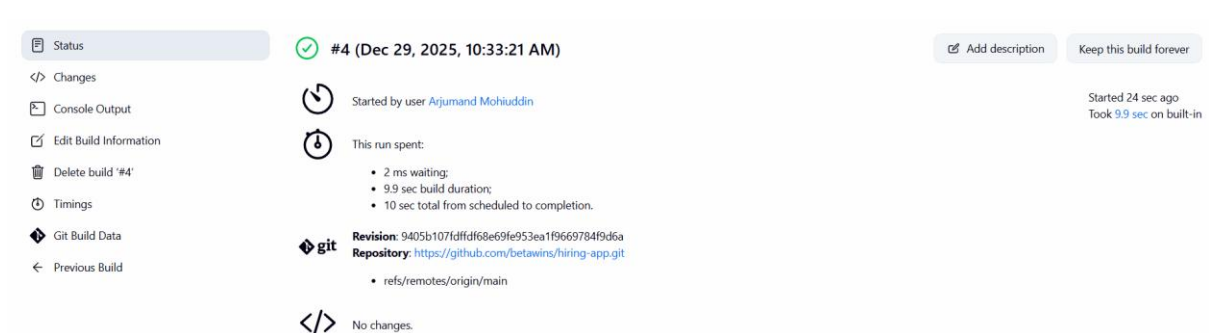
Add details of the project properties along with the token and URL of the sonarqube server



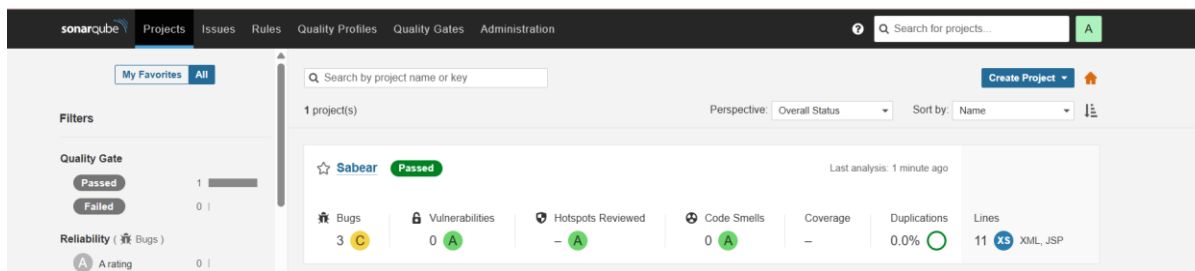
Afer changing the configuration of the job, Sonarqube logo will be appeared, next step will be testing the job the job



Sonarqube job is executed sucessfully

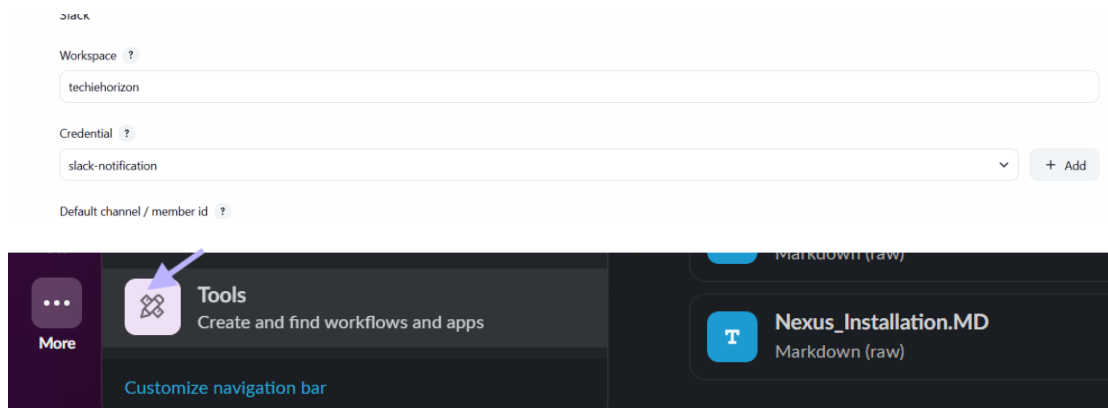


As we can see code quality is passed in sonarqube

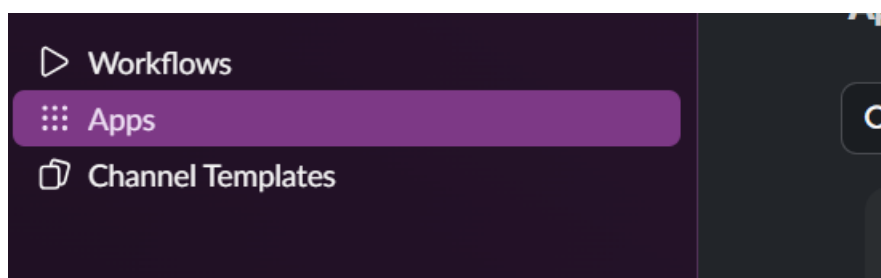


### Stage3: Slack Integration to send the alerts to slack.

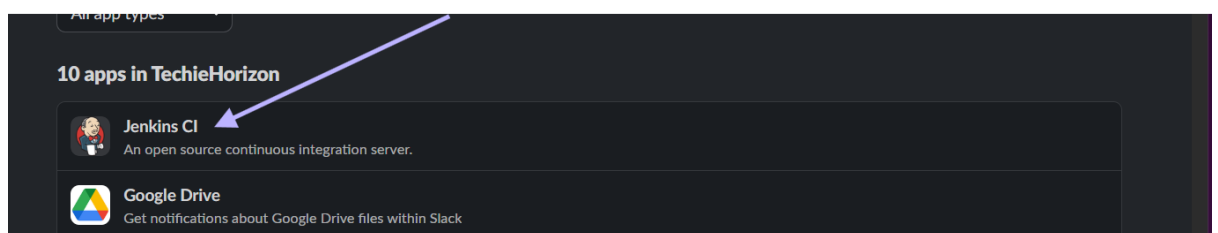
Go to tools in slack and click tools



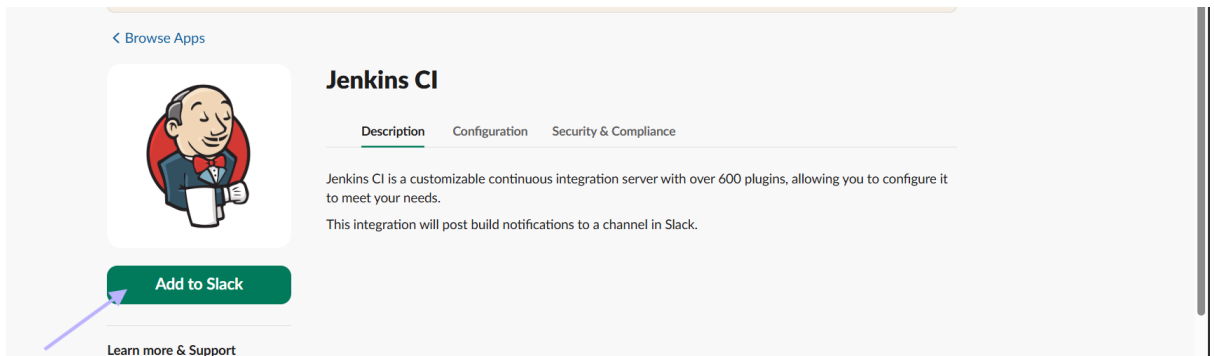
Select apps



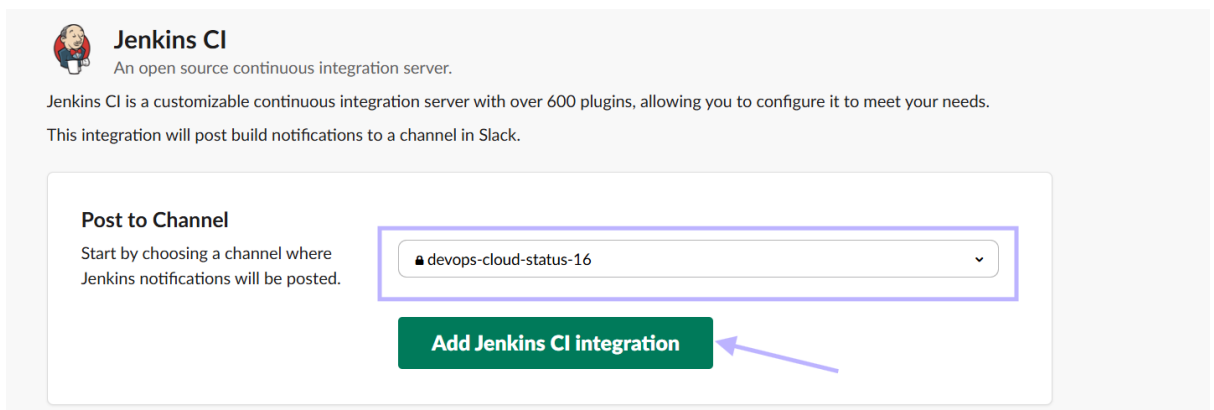
Select Jenkins CI



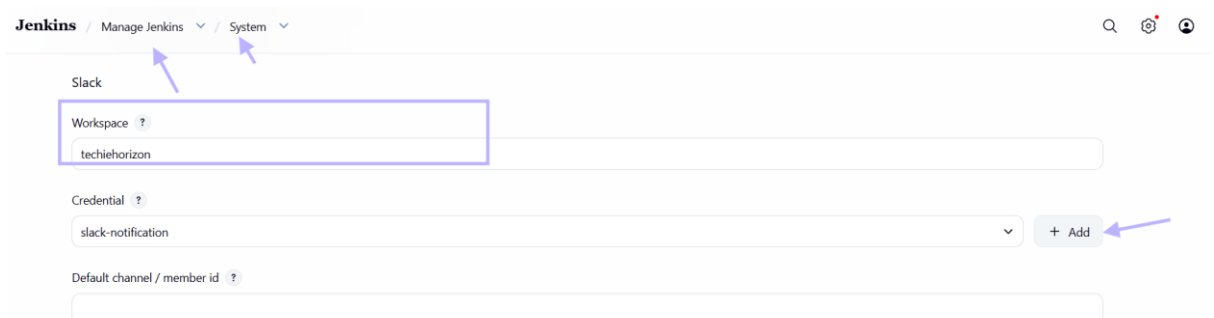
ADD Jenkins CI notifications



## Select the channel

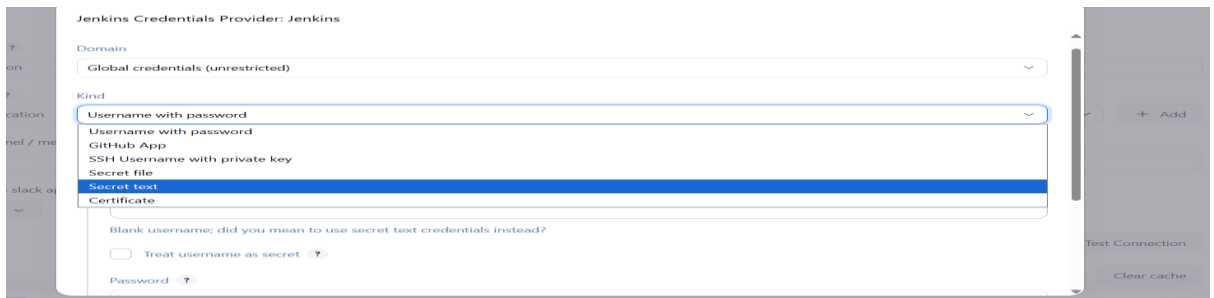


Go to Jenkins manage Jenkins > system configuration > select slack write the name of the organisation then click add credentials



And select secret text





Select token from slacks on step 3

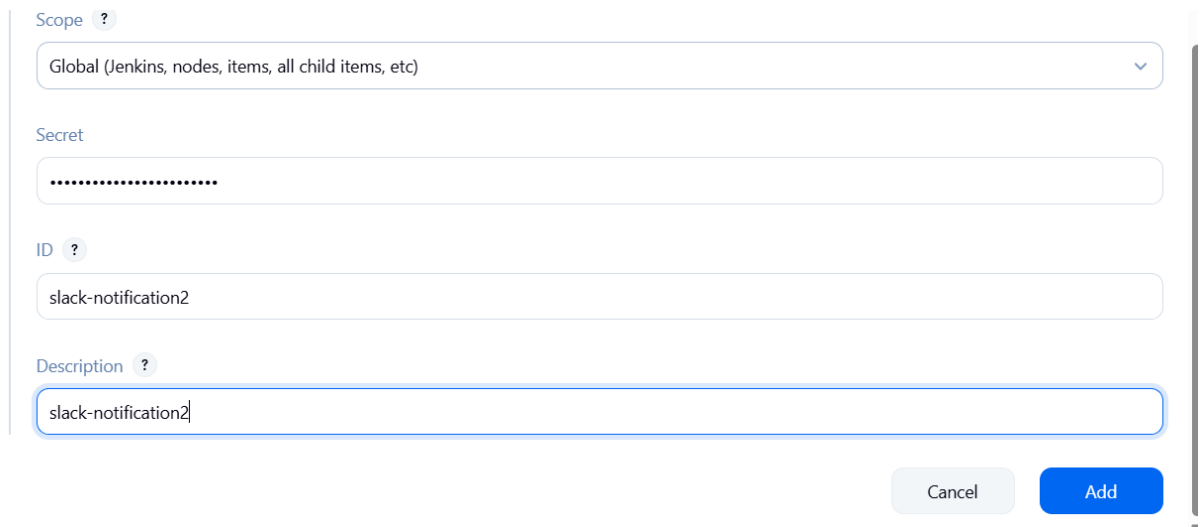
### Step 3

After it's installed, click on **Manage Jenkins** again in the left navigation, and then go to **Configure System**. Find the **Global Slack Notifier Settings** section and add the following values:

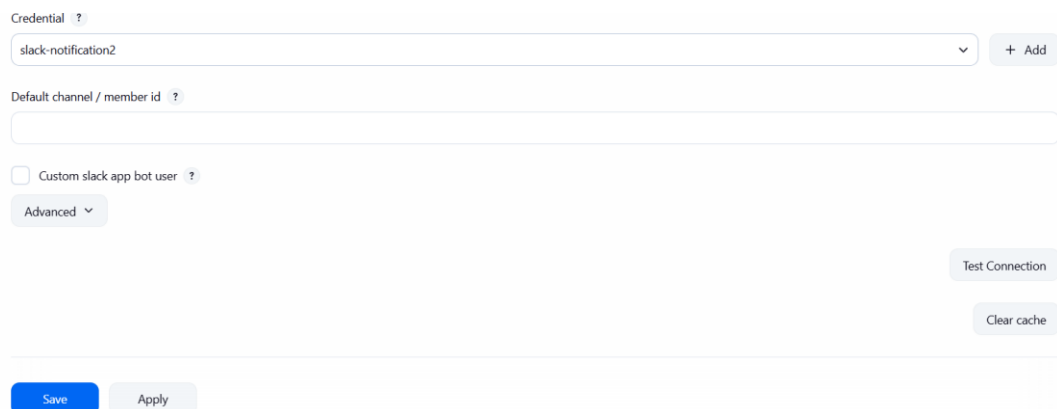
- **Team Subdomain:** `techiehorizon`
- **Integration Token Credential ID:** Create a secret text credential using `yAYWa4TWQKF-fWp7ppazQMBwr` as the value

The other fields are optional. You can click on the question mark icons next to them for more

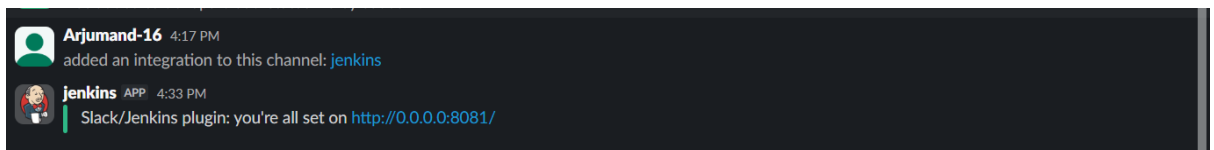
Paste the token and prove ID and description



After adding test connection then save



Tested successfully



### **Error 1: Branch Not Found**

- Issue: Jenkins build failed with “Couldn't find any revision to build”
- Cause: Git repository used main branch instead of master
- Fix: Updated branch specifier in Jenkins from \*/master to \*/main

### **Error 2: SonarQube Authentication Failure**

- Issue: Jenkins could not authenticate with SonarQube
- Cause: SonarQube requires token-based authentication
- Fix: Generated SonarQube token and configured it as a Jenkins Secret

### **Text credential**

### **Error 3: Java Version Compatibility**

- Issue: SonarQube Scanner requires Java 17
- Cause: Default Java version was incompatible
- Fix: Explicitly configured JAVA\_HOME and PATH to use Amazon Corretto Java 17 in Jenkins

build step

### **Error 4: No Build Notifications**

- Fix: Explicitly configured JAVA\_HOME and PATH to use Amazon Corretto Java 17 in Jenkins

build step

### **Error 4: No Build Notifications**

- Issue: No visibility of build results
- Cause: Notifications not configured
- Fix: Configured Slack Notifications under Jenkins Post-build Actions

### **Conclusion**

A complete CI pipeline was successfully implemented using Jenkins integrated with SonarQube and GitHub. SonarQube was deployed using Docker, enabling scalable and isolated code quality analysis.

Jenkins was configured to perform automated code scanning during builds, and Quality Gate results were validated successfully. Slack notifications were enabled to provide real-time build status updates. This setup ensures improved code quality, faster feedback, and better collaboration within the development workflow